

Deep Typographical Layout and Vector Generation in MoonBit

Abstract

This paper presents the design and implementation of **moon-typst**, a zero-dependency typographical layout engine and multi-format vector compiler implemented in MoonBit. It supports **mathematical layout**, two-dimensional mathematical layout including matrices $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$, vectors $\vec{v}$, and limits $\lim_{x \rightarrow \infty}$, and high-fidelity output serialization to SVG, HTML5 and Adobe PDF 1.4.

1. Mathematical Kinetics and Tensor Formulation

Modern computational typesetting requires rigorous spatial bounding algorithms for complex mathematical expressions. This section details the tensor formulation of layout constraints:

When projected into high-dimensional vector spaces $\mathbb{R}^n$, state vectors satisfy the continuity condition $\nabla \cdot \mathbf{J} = 0$:

The asymptotic energy dispersion across frequency bands is evaluated via the limit series $\lim_{I \rightarrow \infty} \frac{1}{I} \sum_{n=1}^I x_n$:

With localized spectral estimation modulated by the accented field estimators $\hat{\theta}$ and $\tilde{\omega}$, baseline coherence is strictly maintained.

2. Implementation in Pure MoonBit

The compiler architecture is built upon deterministic recursive descent with zero dynamic FFI calls:

```
// Geometric layout box dispatch in MoonBit
fn layout_element(ctx : StyleContext, box : LayoutBox) -> Unit {
  let x = box.rect.x
  let y = box.rect.y
  let w = box.rect.w
  let h = box.rect.h
  // Compute baseline alignment and font metrics
}
```

3. Multi-Target Rendering Pipelines

The compilation pipeline provides deterministic generation across three distinct vector targets:

SVG 1.1 Vector Graphics

Scalable XML graphics with font family definitions and path outlines

Adobe PDF 1.4 Binary Document

Self-contained indirect objects, xref tables, and binary font dictionaries

Semantic HTML5

Clean document hierarchy with responsive styling and CSS grid compatibility

4. Benchmark and Evaluation

The benchmark in `benchmarks/pipeline_test.mbt` measures parse, layout, HTML, SVG and PDF stages on a fixed input. Run it for each target under recorded host and toolchain conditions before